

REVIEW ARTICLE

Foundation Models for Neural Signal Decoding: EEG-Centered Perspectives Toward Unified Representations

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ABSTRACT

Neural signals such as EEG, ECoG, and intracortical recordings offer a valuable window into brain dynamics but remain difficult to decode due to high dimensionality, nonstationarity, and substantial interindividual variability. Traditional machine learning and deep learning models often show limited generalizability and insufficient interpretability in these settings. Foundation models (FMs)—large-scale architectures pretrained on diverse datasets—have recently emerged as a promising paradigm for building robust, transferable, and physiologically grounded neural representations. Among these modalities, EEG currently serves as the most practical and representative platform for FM development due to its large-scale open datasets, standardized protocols, and broad clinical applicability, while the same conceptual framework remains generalizable to other neural recording types. This review synthesizes emerging FM approaches for neural decoding and critically examines representative EEG-based architectures. We highlight three essential design principles: physiology-aware representation learning that captures oscillatory and dynamic structure, structure-aware architectures that incorporate spatial and anatomical priors, and interpretability mechanisms that ensure neuroscientific and clinical validity. Although models such as the Patched Brain Transformer, CBraMod, and BrainGPT demonstrate encouraging adaptability, many still inherit objectives from non-neural domains and underutilize spatial priors such as electrode topology or functional connectivity. While this review focuses on EEG as the most data-rich and scalable testbed, the same framework can extend to ECoG and intracortical recordings to support unified neural representations across spatial and temporal scales. Fully realizing the potential of neural FMs will require biologically informed objectives, structure-aware architectures, interpretable representations, and standardized data ecosystems.

Abbreviations: ACPE, Adaptive Criss-Cross Patch Encoding; BCI, brain–computer interface; BIDS, Brain Imaging Data Structure; BrainGPT, Brain Generative Pretrained Transformer; CBraMod, Criss-Cross Brain Model; CE, Conformité Européenne; CNN, convolutional neural network; DL, deep learning; ECoG, electrocorticography; EEG, electroencephalography; EGA, EEG Graph Adapter; ERP, event-related potential; FDA, US Food and Drug Administration; FM, foundation model; GNN, graph neural network; LoRA, Low-Rank Adaptation; LRP, Layer-wise Relevance Propagation; MDR, Medical Device Regulation; MEA, microelectrode array; ML, machine learning; MNI, Montreal Neurological Institute; NLP, natural language processing; PAC, phase–amplitude coupling; PBT, Patched Brain Transformer; RSA, representational similarity analysis; SaMD, Software as a Medical Device; sEEG, stereoelectroencephalography; SNN, spiking neural network; SNR, signal-to-noise ratio; SSL, self-supervised learning; STTransformer, Spatio-Temporal Transformer; ViT, vision transformer.

1 | Introduction

Neural decoding refers to the process of translating brain activity into interpretable behavioral, cognitive, or clinical information (Buzsaki and Draguhn 2004; Wang 2010). It serves as a core computational challenge in both neuroscience (Lewis et al. 2024; Moon et al. 2024) and brain–computer interface (BCI) research (Mak and Wolpaw 2009; de Peralta et al. 2007; Shende and Jabade 2015), aiming to infer latent mental (Lehembre et al. 2012; Morlet et al. 2023; Choe et al. 2025) or physiological states from measurable neural activity patterns (Staba et al. 2014; Tripathi et al. 2010). Achieving reliable decoding requires models that can generalize across subjects, tasks, and recording conditions, while maintaining sensitivity to biologically meaningful features. A variety of neural recording modalities—such as electroencephalography (EEG), electrocorticography (ECoG), and intracortical microelectrode arrays (MEAs)—provide the data foundation for decoding these dynamics, each offering distinct trade-offs in spatial and temporal resolution (Buzsáki et al. 2012; Buzsáki 2004; Waldert 2016). Recent advances in neural signal acquisition and computational modeling have accelerated efforts toward robust neural decoding (Simeral et al. 2021; Hayati 2025); however, traditional machine learning (ML) and deep learning (DL) approaches often struggle to maintain performance across diverse datasets and conditions (Xu et al. 2020). This limitation motivates the exploration of foundation models (FMs) as a unifying paradigm for scalable, generalizable, and physiologically grounded neural decoding (Zhou et al. 2025; Cui et al. 2024).

FMs—large-scale neural networks pretrained on extensive and diverse datasets—represent a paradigm shift in representation learning (Bommasani 2021) (Figure 1). Originally developed

in natural language processing (NLP) (Myers et al. 2024; Paaß and Giesselbach 2023) and computer vision tasks (Yuan et al. 2021; Yang et al. 2025), exemplified by models such as GPT for language (Brown et al. 2020) and UNI for pathology images (Chen, Ding, et al. 2024), FMs differ fundamentally from conventional large-scale DL systems not only in scale but in their two-stage learning paradigm (Bommasani 2021). Specifically, FMs are first pretrained on broad, unlabeled data to capture general, transferable representations, and are subsequently adapted or fine-tuned for downstream tasks across diverse domains (Yuan 2023). This paradigm enables cross-domain generalization and adaptation, allowing FMs to learn modality-agnostic representations that can flexibly transfer to new contexts (Dwivedi et al. 2024; Wang, Fan, et al. 2025). When applied to neural signals, such models potentially offer improved generalization across individuals and recording sessions, robustness to noise and intersubject variability, multitasking capabilities, and adaptability to heterogeneous neural input formats (Wang et al. 2025; Klein et al. 2025; Wang, Zhao, et al. 2024).

Recent models, such as Patched Brain Transformer (PBT) (Klein et al. 2025), Criss-Cross Brain Model (CBraMod) (Wang, Zhao, et al. 2024), and Brain Generative Pretrained Transformer (BrainGPT) (Yue et al. 2024), have demonstrated the feasibility of applying FM architectures to neural decoding tasks. However, significant challenges persist. Current FM frameworks are often direct adaptations from other domains like language or vision, failing to adequately address the unique biophysical properties of neural signals. Critical issues include (1) the mismatch between conventional pretraining objectives and the nonstationary, oscillatory dynamics of brain activity, (2) the failure to incorporate crucial neuroanatomical priors—such as electrode topology and functional connectivity—into

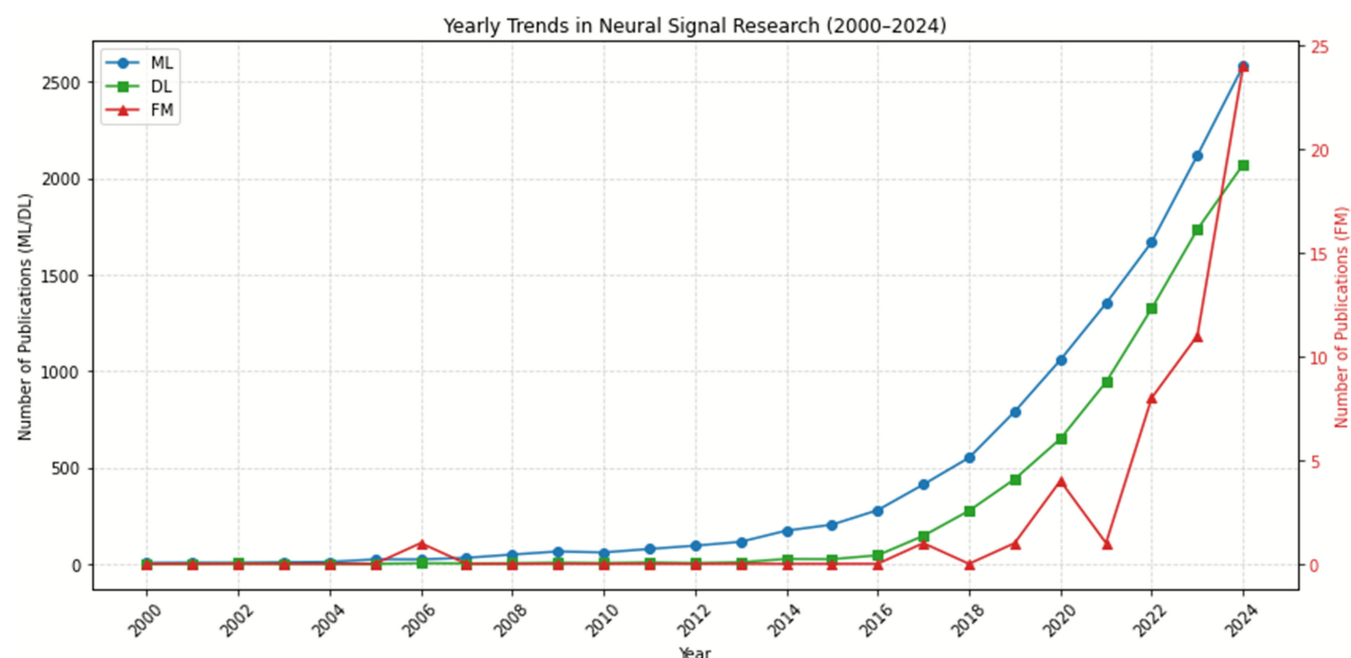


FIGURE 1 | Yearly trends in neural signal research publications from 2000 to 2025, comparing machine learning (ML), deep learning (DL), and foundation model (FM) approaches.

the model architecture, and (3) the lack of principled interpretability frameworks to ensure that model representations are not just performant but also neuroscientifically valid and clinically trustworthy.

Among various neural recording modalities, EEG provides the most practical and representative testbed for developing and validating FMs. Unlike invasive modalities such as ECoG or stereo EEG (sEEG), EEG benefits from standardized acquisition protocols, broad clinical and experimental applicability, and extensive open-access datasets that enable large-scale pretraining and benchmarking at FM scale. These advantages make EEG uniquely suited for early-stage FM research, allowing systematic exploration of pretraining objectives, model architectures, and cross-subject generalization before extending such frameworks to more specialized modalities. Therefore, this review primarily focuses on EEG-based FMs while situating the discussion within a broader, modality-general framework toward unified neural representations.

This review critically examines the role and potential of FMs in brain signal decoding. We begin by analyzing the limitations of traditional ML/DL approaches, illustrating why FMs uniquely address these challenges. Next, we explore essential model-design considerations specific to neural data, structured around three core dimensions: (1) physiology-aware representation learning, (2) structure-aware model architecture, and (3) mechanisms to enhance interpretability and facilitate clinical translation. We then evaluate recent FM implementations in this context and conclude by outlining open challenges and future research directions.

2 | Methods—Literature Review Strategy

To ensure a comprehensive and unbiased synthesis of the literature, a structured review process was conducted to identify relevant studies on FMs for neural signal decoding. The review methodology followed established principles of systematic literature analysis, adapted for interdisciplinary topics that span neuroscience, artificial intelligence, and biomedical engineering.

2.1 | Databases and Search Scope

The literature search was performed using three major databases widely used in neuroscience and computational modeling research: PubMed, Google Scholar, arXiv, and IEEE Xplore. The search covered publications from January 2019 to October 2025, corresponding to the period during which FMs and self-supervised learning emerged as dominant paradigms in the ML community. This time frame also captures the initial appearance of FM-related studies applied to EEG and other neural modalities. Notably, arXiv preprints were also included because the field of DL-based FMs is rapidly evolving, and many influential FM studies are first disseminated as preprints prior to journal publication. Excluding arXiv would therefore overlook a substantial portion of state-of-the-art methods relevant to neural decoding.

2.2 | Search Keywords and Query Formulation

To capture studies at the intersection of FM development and neural signal decoding, a combination of controlled vocabulary and free-text keywords was used. The core query included terms such as “foundation model,” “large-scale model,” “self-supervised learning,” “transformer,” and “representation learning,” combined with neural decoding-related terms including “EEG,” “ECoG,” “intracortical recording,” “neural decoding,” and “BCI.” Boolean operators were applied to each database syntax to ensure consistent retrieval. This strategy enabled inclusion of both general-purpose FM frameworks and modality-specific adaptations relevant to brain signal analysis.

2.3 | Inclusion and Exclusion Criteria

Studies were included if they were peer-reviewed publications, conference proceedings, or preprints (e.g., arXiv) published between 2019 and 2025 that proposed, evaluated, or analyzed FM-based architectures for neural signal decoding. Preprints were included because the field of DL-based FMs evolves rapidly, and many influential FM studies are initially released as arXiv manuscripts prior to journal publication. Eligible works were required to provide methodological details, training strategies, or conceptual contributions that connect large-scale pretraining to neural data. Studies were excluded if they focused solely on traditional DL models without pretraining or transfer learning, lacked methodological detail, or used non-biological simulation datasets. Non-English publications and brief abstracts without peer review were also excluded to ensure consistency and scientific rigor.

2.4 | Study Selection Process

All retrieved records were screened independently by two authors to ensure relevance and quality. Titles and abstracts were first reviewed to filter out irrelevant studies, and full texts of potentially eligible papers were then assessed for inclusion. Duplicates across databases were removed manually. From the remaining pool, key studies were prioritized based on their methodological novelty, impact, and direct relevance to the three thematic pillars of this review—physiology-aware representation learning, structure-aware model architecture, and interpretability for clinical translation. This two-stage selection ensured both breadth and depth in coverage.

2.5 | Data Extraction and Synthesis

For each selected study, we extracted information regarding model architecture, pretraining objectives, dataset modality, evaluation metrics, and interpretability methods. The extracted data were synthesized narratively to identify converging trends and gaps in the current literature. When multiple studies shared similar methodological foundations, representative works were highlighted to illustrate the evolution of FM approaches for neural decoding. Findings were ultimately organized according to the strategic framework proposed in this review, emphasizing

the interplay between physiological grounding, structural modeling, and clinical applicability.

3 | Challenges in Neural Decoding From Neural Signals

3.1 | Limitations of Traditional ML and DL Approaches

While ML and DL methods have demonstrated strong performance in controlled, task-specific settings—e.g., seizure detection (Yuan et al. 2018; Ullah et al. 2018), motor intention decoding (Tabar and Halici 2016; Dai et al. 2019), sleep stage classification (Li et al. 2022; Eldele et al. 2021), and mental state recognition (Zeng et al. 2018; Wang et al. 2014)—they often fail to scale beyond those narrow domains. This limitation stems from a number of fundamental assumptions embedded in traditional approaches (Vallabhaneni et al. 2021; Tortora et al. 2020).

Most conventional models are trained using supervised learning on carefully curated, labeled datasets, often sourced from a single subject, session, or hardware configuration (Zhang et al. 2024; Han and Jeong 2021; Roy et al. 2019; Chen et al. 2022). These models generally assume that training and test data are drawn from similar distributions. However, in real-world neural decoding applications, even minor changes in experimental design, task protocol, or recording quality can lead to substantial performance degradation (Jayaram and Barachant 2018; Lotte et al. 2018).

Moreover, these models show limited generalization across hardware systems and sensor configurations (Bethge 2022; Han et al. 2023). ML models trained on EEG data from one device or sampling frequency often require retraining or extensive preprocessing to function in another setup. The problem is even more pronounced across modalities. Although EEG and ECoG may capture similar underlying neural dynamics, differences in spatial resolution, signal-to-noise ratio (SNR), electrode placement, and invasiveness introduce substantial domain shifts, making it difficult to transfer models across modalities without performance loss (Graitmann 2005; Peterson et al. 2021; Desai et al. 2019). Collectively, these signal- and system-level variations undermine model generalization, often necessitating retraining for each subject, session, or configuration (Han and Jeong 2021; Han et al. 2023; Wu et al. 2020).

Another critical limitation lies in interpretability. Deep neural networks, such as convolutional or recurrent architectures, capture complex, nonlinear patterns but often do so through distributed representations that are difficult to map onto known neural processes or anatomical structures. This “black-box” nature impedes their adoption in scientific and clinical domains where explainability and reliability are essential (Torres et al. 2023; Gurmessa and Jimma 2025; Sturm et al. 2016; Sujatha Ravindran and Contreras-Vidal 2023). While methods such as saliency maps or attention mechanisms have been proposed to address this, they remain

preliminary and often lack neurobiological validity (Farahat et al. 2019; Shimizu et al. 2024).

In sum, while ML/DL models have shown potential in controlled environments, their dependence on homogenous, labeled data, lack of robustness to signal variability, and poor interpretability highlight the need for a more flexible and scalable modeling framework—one that can effectively incorporate the nonlinear, nonstationary, and high-dimensional nature of neural data while facilitating knowledge transfer and preserving neurobiological insight.

4 | FMs for Neural Signal Decoding: Opportunities and Requirements

The emergence of FMs—large-scale neural architectures pretrained on vast and diverse datasets—has transformed representation learning in domains such as NLP (Myers et al. 2024; Paaß and Giesselbach 2023; Achiam et al. 2023), computer vision (Yuan et al. 2021; Yang et al. 2025; Chen, Ding, et al. 2024), speech processing (Défossez et al. 2024; Ku et al. 2025; Wang et al. 2023; Guo et al. 2024) and later extended to multimodal frameworks that integrate information across text, image, and audio modalities (Liu et al. 2024; Stevens 2024; Wu 2024; Xu et al. 2023). These models excel in learning generalizable, transferrable, and task-agnostic representations, enabling rapid adaptation to downstream tasks with minimal or even no supervision (Bommasani 2021; Narayan et al. 2022; Awais et al. 2025). Unlike traditional models that are trained in a task-specific, narrowly scoped fashion, FMs are designed to learn from heterogeneous, unlabeled, and multiscale data through unsupervised or self-supervised objectives, constituting a general-purpose learning strategy that departs fundamentally from task-specific supervised learning.

While FMs have been rapidly adopted in structurally consistent domains such as MRI, where anatomical variability is relatively constrained, their application to neural data has progressed more slowly (Barbano et al. 2024; Tak et al. 2024; Cox et al. 2024; Zhang et al. 2025; Sun et al. 2025). This is largely due to the substantial interindividual variability and signal-level noise inherent in neural recordings, which complicate the learning of generalizable representations (Xu et al. 2020; Cui et al. 2024; Gundu and Panem 2025; Aristimunha et al. 2025; Turgut et al. 2025; Keene et al. 2000). Nonetheless, this very setting—characterized by limited, noisy labels but abundant unlabeled data—makes neural signal analysis a particularly suitable domain for FM-based approaches (Zhou et al. 2025). The integration of SSL with large-scale datasets is now bridging this gap, as SSL enables models to learn intrinsic temporal and spectral structures directly from unlabeled neural signals. By leveraging objectives such as masked reconstruction, contrastive alignment, or autoregressive forecasting, FMs can extract domain-agnostic representations that generalize across subjects and recording conditions. Combined with the growing availability of large, diverse neural datasets, these developments are fostering scalable and flexible models capable of capturing complex, multiscale neural dynamics beyond narrow, task-specific objectives (Weng et al. 2024;

Truong et al. 2024; Agarwal et al. 2022; Gramfort 2021; Del Pup et al. 2023).

4.1 | Emerging FMs in Neural Signal Decoding

Recent studies exploring structural, topographical, and graph-based representations reflect the growing interest and ongoing efforts in applying FM frameworks to neural data (Wang, Suzumura, and Kanezashi 2024; Suzumura et al. 2024). Additionally, breakthroughs in attention-based architectures for time-series analysis—such as transformers and their adaptations for long-range temporal modeling—have significantly improved the ability to capture complex, hierarchical dependencies in neural activity (Klein et al. 2025; Wang, Zhao, et al. 2024; Shi et al. 2024; Chen, Ren, et al. 2024; Aksan et al. 2021; Cui 2024). Combined with the increasing availability of large-scale neural datasets, these developments have brought EEG-based FMs closer to practical realization (Da Silva et al. 2024; Obeid and Picone 2016; Goldberger et al. 2000). Building on this technological foundation, several recent models have applied FM paradigms to neural signal decoding. These models vary in architectural choices, pretraining strategies, and representation goals, reflecting different interpretations of how FM principles can be adapted to the unique challenges of neural signal decoding.

Recent EEG FMs can be categorized based on the type of inductive biases they incorporate, such as spectro-temporal structure, spatial graph topology, intersubject variability, and generative predictability. These models primarily adopt self-supervised learning objectives to capture generalizable representations from raw EEG data, aiming to overcome challenges posed by high signal variability, noise, and limited labeled data.

One line of approaches leverages the spectro-temporal characteristics of EEG signals by modeling complex interactions across both frequency and time domains (Aksan et al. 2021; Jiang, Zhao, and Lu 2024). A representative example is the Spatio-Temporal Transformer (STTransformer), which is designed to jointly model the spectral and temporal characteristics of EEG signals. By incorporating multiresolution spectral features and applying attention mechanisms across both frequency and time dimensions, this model captures rich cross-band and cross-time dependencies. Such joint modeling directly enhances decoding accuracy by allowing the model to recognize oscillatory patterns that correspond to cognitive or motor states. Trained using a masked modeling objective on spectro-temporal patches, the STTransformer demonstrates the potential of adapting self-supervised token-based learning to the unique structure of EEG signals.

Another direction focuses on representing the spatial structure of EEG sensors using graph-based methods (Yue et al. 2024; Wang, Suzumura, and Kanezashi 2024; Suzumura et al. 2024). Another notable approach is EEG-Graph Adapter (EGA), which captures the structural characteristics of brain organization by modeling EEG signals as graph-structured data. EGA introduces a lightweight graph neural network (GNN)-based adapter to learn spatial dependencies among

EEG sensors. By integrating spatial priors into the temporal backbone, EGA improves decoding robustness under heterogeneous electrode layouts and facilitates spatially interpretable feature extraction—critical for clinically reliable decoding. The model also achieves efficient adaptation by freezing the pretrained temporal backbone and fine-tuning only the GNN adapter, thereby reducing computational overhead and data requirements.

Some models focus more directly on the temporal aspect of EEG by segmenting the signal into short patches and applying masked modeling strategies similar to those used in NLP (Klein et al. 2025; Shi et al. 2024; Chen, Ren, et al. 2024). PBT follows this paradigm by training a transformer encoder to predict masked segments within the time series, thereby learning temporal dependencies in a scalable and unsupervised manner. This patch-based pretraining improves decoding generalization across tasks by enabling robust temporal representation learning even under limited labeled data.

To address variability across individuals, contrastive learning has also been explored as a way to enforce consistent representations across subjects (Wang, Zhao, et al. 2024). CBraMod adopts this strategy by aligning latent representations through a contrastive objective, promoting generalization in population-level decoding tasks. This directly mitigates one of the most persistent decoding challenges—subject-to-subject variability—by learning invariant representations that remain discriminative for clinically relevant signal differences.

Finally, generative models inspired by autoregressive language modeling offer an alternative approach by learning to predict the next EEG token in a sequence (Yue et al. 2024; Wang, Huang, et al. 2025; Jiang, Wang, et al. 2024). BrainGPT is a notable example that tokenizes EEG signals and uses a unidirectional transformer trained to generate future tokens (Yue et al. 2024). This autoregressive mechanism supports predictive decoding by leveraging temporal continuity and forecasting future neural dynamics, which is valuable for real-time BCI control or seizure anticipation. This enables applications such as signal forecasting, inpainting, and real-time adaptation in closed-loop BCIs (Table 1).

Across these EEG FMs, each approach demonstrates distinct advantages and trade-offs. PBT achieves high parameter efficiency through a vision transformer (ViT)-style patch encoder but remains limited by its reliance on task-specific supervised pretraining, which restricts generalization across diverse paradigms. CBraMod effectively captures both spatial and temporal dependencies using criss-cross attention and masked reconstruction, while its contrastive learning strategy promotes intersubject alignment. However, this approach may inadvertently suppress subject-specific neural idiosyncrasies that carry clinically meaningful information, highlighting the need for a balance between population-level harmonization and individual-level specificity. BrainGPT, in contrast, employs an autoregressive transformer that excels in modeling long-range temporal continuity and supports generative tasks such as signal forecasting. Nonetheless, it currently lacks physiology-aware learning objectives (e.g., phase-amplitude coupling [PAC] or oscillatory-band regularization) and does not include

TABLE 1 | Comparison of three recent EEG foundation models (architectural characteristics, pretraining strategies, and alignment with strategic design directions for neural decoding).

Model	Backbone architecture	Pretrain strategy	Strengths	Physiology-aware representation learning	Structure-aware architecture	Interpretability and clinical robustness
PBT	ViT-style transformer with linear patch embedding	Supervised pretraining with data augmentation (motor imagery)	High efficiency and strong decoding performance with few parameters	△ Supervised pretraining only, lacks biological or oscillatory self-supervised objectives	△ Patch-wise input with positional encoding; flexible input length but lacks explicit topographic modeling	✗ No interpretability methods; not designed for clinical robustness
CBraMod	Criss-cross transformer (spatial + temporal attention)	Self-supervised masked EEG reconstruction (patch-based, time + frequency)	Adaptability to heterogeneous EEG formats and large dataset evaluation	✓ Time + frequency domain patch encoding; masked reconstruction capturing oscillatory structure	✓ Criss-cross attention for spatial and temporal disentanglement; Adaptive Criss-Cross Patch Encoding (ACPE) for variable formats	✗ No interpretability methods; limited clinical insight
BrainGPT	Autoregressive transformer with electrode-wise embedding and causal self-attention	Next-token prediction (autoregressive forecasting)	Strong temporal modeling enabling predictive and generative decoding (e.g., signal forecasting)	✗ No physiology-aware or neurobiologically grounded objectives	△ Electrode-wise embedding but limited modeling of cortical geometry and topological structure	✗ No interpretability methods; lacks clinical robustness

Note: Symbols: ✓ = addressed; △ = partially addressed; ✗ = not addressed.

explicit interpretability or clinical robustness mechanisms. Together, these models illustrate how FM architectures can enhance neural decoding performance through improved temporal modeling, cross-subject generalization, and generative prediction, while also revealing remaining challenges in physiological grounding and interpretability.

4.2 | Challenges for Applying FMs

While several recent models have demonstrated the technical feasibility of applying FM-like components to neural data—incorporating attention mechanisms, self-supervised objectives, and neurophysiologically informed designs such as graph representations and frequency-aware modeling—the direct adaptation of these frameworks to neural signals remains challenging. This is largely due to a fundamental mismatch between the architectural assumptions of existing FMs and the unique biophysical properties of neural signals, which are nonstationary, noisy, and exhibit substantial interindividual variability (Xu et al. 2020; Aristimunha et al. 2025; Turgut et al. 2025; Keene et al. 2000).

4.2.1 | Mismatch Between FM Objectives and Neurophysiological Structure

A primary challenge lies in the adoption of pretraining tasks that are conceptually and computationally derived from domains such as language (e.g., masked language modeling) or vision (e.g., contrastive patch alignment). These objectives assume discrete, symbolic, or spatially localized structures, whereas neural signals such as EEG and ECoG are continuous, nonstationary, and shaped by complex oscillatory and nonlinear dynamics. Current models, including STTranformer and PBT, utilize patch-based masking objectives that may successfully learn general dependencies, but do not explicitly capture biologically meaningful constructs such as PAC, interchannel coherence, or multiscale neural rhythms (Grigorovsky et al. 2020; Onslow et al. 2011; Samiee and Baillet 2017; Lopez-Bernal et al. 2024; Shin et al. 2021; Shin et al. 2022). Beyond oscillatory phenomena, biologically relevant characteristics of neural signals also include event-related potentials (ERPs), transient evoked responses, neuronal spiking statistics, and network-level synchronization patterns. Future FM objectives should therefore encompass both oscillatory and non-oscillatory signal properties to achieve more comprehensive physiological representation learning.

4.2.2 | Insufficient Incorporation of Structural Priors in Architecture Design

Beyond training objectives, many current FM-based models fail to incorporate the spatial and anatomical priors inherent in neural data acquisition. Neural signals, for example, are recorded via a fixed grid of electrodes with known spatial relationships governed by cranial anatomy. Yet, standard transformers and convolutional architectures treat EEG channels as independent or permutation-invariant inputs, ignoring spatial coherence and functional connectivity patterns.

Although recent models such as EGA have begun incorporating spatial priors via GNN, their implementations remain shallow and anatomically constrained. Specifically, EGA captures only fixed anatomical adjacency based on electrode positioning, overlooking deeper structural properties of the brain such as the hierarchical organization of cortical regions, interhemispheric symmetry, and spatial modularity—features that play a critical role in large-scale brain dynamics. Additionally, by freezing the temporal backbone and fine-tuning only a lightweight graph adapter, its capacity to learn rich spatio-temporal patterns and adapt to interindividual variability is inherently limited (Priyasat et al. 2022; Yang et al. 2020) (Table 1).

4.2.3 | The Interpretability Gap and Neuroscientific Validation

Interpretability remains a critical bottleneck for the adoption of FM-based neural decoders in both scientific and clinical contexts. Although tools such as attention maps and saliency visualizations provide preliminary insights into model behavior, these explanations often remain superficial and lack neuroscientific grounding (Thomas et al. 2019; Chefer et al. 2021; Doshi-Velez and Kim 2017). For instance, high attention weights assigned to certain spectro-temporal features or graph nodes do not necessarily reflect biologically meaningful constructs such as ERPs, neural oscillations, or functional brain networks. Moreover, the absence of systematic validation linking learned representations to experimentally verifiable neural signatures limits their interpretive value. Developing principled interpretability frameworks that align model-inferred representations with measurable neural mechanisms—such as stimulus-locked responses, cortical phase synchronization, or connectivity gradients—will be essential for ensuring neuroscientific and clinical reliability (Hutson 2018; Rudin 2019).

4.3 | Comparative Analysis of Pretraining Strategies in EEG FMs

Recent EEG FMs have largely adapted pretraining strategies originally designed for natural language and vision domains—such as masked token prediction, contrastive learning, or autoregressive next-token generation. While these objectives have proven effective for learning contextual or spatial relationships in symbolic or image-based data, they are not inherently aligned with the continuous, nonstationary, and oscillatory dynamics of neural signals. EEG and related electrophysiological data possess temporal-spectral structures that differ fundamentally from the syntactic and spatial dependencies that underpin NLP and vision models. Consequently, current EEG FMs capture only a subset of the underlying neural information space, leaving many physiologically meaningful patterns underrepresented.

From a comparative perspective, PBT focuses on temporal dependency learning through masked segment prediction, effectively modeling local continuity within time-series windows. However, because its supervision is limited to sequential reconstruction, it struggles to encode cross-frequency coupling and long-range coherence that are central to neural communication. CBraMod, on the other hand, combines

spatial and temporal attention with contrastive learning objectives to achieve population-level alignment across subjects. This facilitates robustness to individual variability but introduces the risk of over-normalization, where the model suppresses subject-specific neurophysiological signatures—potentially discarding subtle yet clinically relevant biomarkers. BrainGPT, as an autoregressive transformer, represents an ambitious step toward generative neural modeling, successfully capturing long-range temporal dependencies and enabling sequence forecasting. Nonetheless, its next-token objective remains purely statistical and lacks any explicit mechanism to learn oscillatory phase relationships, band-limited synchrony, or nested rhythms that define neurophysiological communication.

A closer examination of these pretraining paradigms reveals that they primarily encode surface-level temporal or spatial correlations rather than deeper neurobiological mechanisms. For example, masked modeling encourages context reconstruction but does not enforce alignment with rhythmic phase dynamics; contrastive alignment maximizes intersubject similarity but may ignore functional asymmetries between cortical regions; and autoregressive prediction optimizes likelihood over signal continuity rather than phase coherence or amplitude–phase interactions. These limitations collectively underscore that present EEG FMs, though innovative, remain “first-generation” frameworks—their representations are powerful for decoding tasks but still detached from the physiological processes that generate the recorded signals.

Bridging this gap requires pretraining strategies that directly incorporate neurophysiological constraints into the learning objective. Such constraints might include predictive modeling of PAC, reconstruction of interchannel coherence maps to capture functional connectivity, or frequency-specific self-supervision schemes that reflect established spectral hierarchies in the human brain (e.g., delta–theta–alpha–beta–gamma bands). Integrating these biologically motivated tasks within large-scale pretraining pipelines could transform EEG FMs from generic sequence learners into physiology-aware neural decoders capable of generalizing across individuals and experimental contexts while maintaining interpretable alignment with cortical dynamics.

5 | Strategic Design Directions for FMs in Neural Decoding

Whereas the previous section addressed why conventional FM-based approaches face limitations, here, we focus on how those limitations can be addressed through actionable design innovations. We organize these strategies around three interrelated dimensions: (1) physiology-aware representation learning, (2) structure-aware model architecture, and (3) interpretability and clinical robustness (Table 2).

We define physiology-aware representation learning as the development of self-supervised or pretraining objectives that reflect the temporal, spectral, and dynamic properties of neural

TABLE 2 | Summary of key design dimensions for foundation models in neural decoding. Each dimension includes representative modeling strategies and their alignment with the underlying physiological and structural properties of neural signals.

Dimensions	Key design considerations	Aligned neural signal properties
Physiology-aware representation learning	- Temporal prediction tasks (e.g., forecasting and masked time-step recovery) - Cross-channel reconstruction (e.g., coherence matrices) - Phase–amplitude or spectral coupling modeling	- Temporal dependencies - Oscillatory and rhythmic structure - Interchannel coordination
Structure-aware model architecture	- Spatially informed mechanisms (e.g., graph encodings of electrode topology) - Multiscale temporal modeling (e.g., wavelet-based and dilated attention) - Frequency-specific branches and cross-frequency fusion - Adaptation to variable electrode layouts - Projection to shared spatial reference frames	- Spatial topography of brain signals - Multiscale neural rhythms - Nonlinear dynamics - Intersubject anatomical variability - Hardware-induced topographical shifts
Interpretability and clinical robustness	- Attention visualization across channels and frequency bands - Saliency-based explanation and LRP-style backpropagation - Uncertainty estimation and missing channel handling	- Need for transparent and clinically trusted outputs - Tolerance to low-SNR or missing data - Relevance to diagnostic reasoning

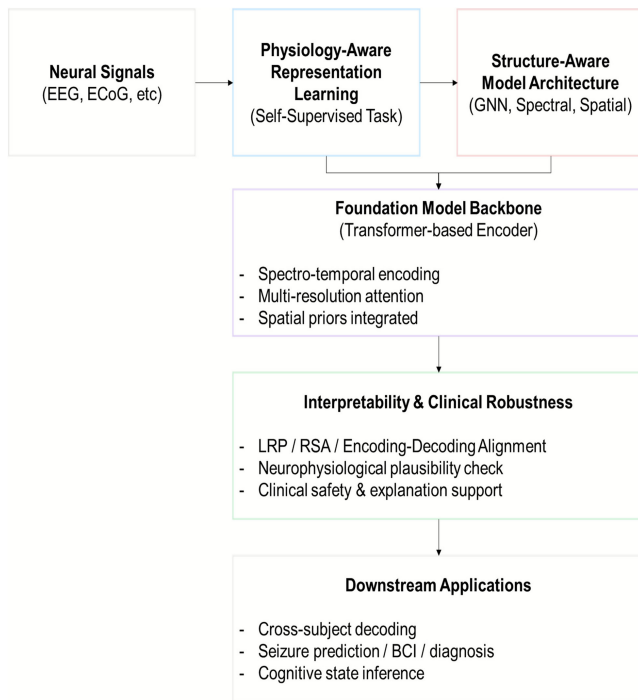


FIGURE 2 | Schematic illustration of the proposed strategic framework for developing foundation models in neural decoding. The process begins with raw neural signals (e.g., EEG and ECoG), which are learned through physiology-aware pretraining tasks such as phase-amplitude coupling reconstruction and temporal forecasting. These representations are then processed through a structure-aware model architecture that integrates anatomical priors (e.g., electrode topology, spatial graphs, or frequency-specific modules). Interpretability is embedded throughout the pipeline using relevance attribution and neuroscientific validation methods. The resulting models support robust and explainable applications in neuroscience and clinical domains.

data. In contrast, structure-aware architecture design refers to modeling approaches that incorporate spatial, anatomical, and topological priors into the structural design of the model itself. These two levels—objectives and architectures—must work in tandem to ensure models learn representations that are both powerful and biologically meaningful (Figure 2).

5.1 | Physiology-Aware Representation Learning

To overcome the representational mismatch between conventional FM objectives and the complex structure of neural signals, future work should prioritize the design of biologically informed SSL strategies. Existing pretraining objectives—such as masked token prediction or patch-level contrastive learning—are conceptually derived from symbolic or spatially localized domains like language and vision. As such, they often fail to capture the continuous, nonstationary, and oscillatory nature of neural data such as EEG and ECoG, as well as the substantial intersubject variability stemming from anatomical, functional, and sensor-related differences. Moreover, these conventional objectives may inadvertently encourage the learning of representations tied to recording artifacts rather than biologically meaningful patterns (Roy et al. 2019; Schirrmeyer et al. 2017; Mumtaz et al. 2021).

To better reflect the intrinsic properties of neural signals, we propose several novel self-supervised tasks grounded in neurophysiological principles. These include PAC prediction to encourage sensitivity to nested oscillatory interactions, interchannel coherence reconstruction to promote modeling of functional connectivity patterns, temporal jigsaw reordering to impose temporal structure learning, cross-modal prediction (e.g., estimating amplitude from phase) to capture latent relationships between signal dimensions, and frequency-aware masking to bias learning toward relevant spectral bands (Banville et al. 2021; Zhang et al. 2022). While recent models such as STTransformer (with masked spectro-temporal prediction) and BrainGPT (with autoregressive forecasting) represent promising steps, existing self-supervised objectives remain largely generic and detached from the biological underpinnings of neural activity. Integrating physiology-aware objectives into large-scale pretraining pipelines offers a promising path forward—potentially enabling models that generalize better across subjects and require less supervision.

To make the concept of physiology-aware representation learning more actionable, we further propose several concrete, biologically grounded self-supervised tasks that operationalize neurophysiological principles within large-scale EEG pretraining. A representative example involves predicting the amplitude envelope of a high-gamma band (80–150 Hz) from the phase of a theta band (4–8 Hz) within the same or a functionally connected cortical region. This PAC prediction task directly reflects hierarchical oscillatory communication, enabling the model to learn cross-frequency dependencies that are fundamental to neural coordination. Similarly, a coherence-based reconstruction objective could train the model to infer interchannel synchrony by reconstructing coherence matrices from partial observations, thereby encouraging the encoding of long-range functional connectivity. Another feasible task, phase-locked temporal reordering, could require the model to restore the correct temporal order of EEG segments based on rhythmic phase information, promoting sensitivity to temporal regularities embedded in oscillatory cycles.

In addition to task-level designs, spiking neural networks (SNNs) represent a promising architecture for implementing physiology-aware learning (Cui et al. 2024; Joshi et al. 2025). Unlike conventional transformer-based models that process continuous-valued signals, SNNs emulate the discrete spike-based communication of biological neurons, providing an intrinsic mechanism for temporal precision and energy-efficient dynamics—both highly relevant for real-time neural decoding and BCIs. While large-scale SNN pretraining remains challenging due to optimization instability and limited dataset standardization, hybrid transformer–SNN frameworks may offer a biologically plausible path toward future EEG FMs that unify energy efficiency, temporal fidelity, and representational interpretability.

5.2 | Structure-Aware Model Architecture

While pretraining objectives determine what a model learns, the architecture governs how neural signals are processed and represented. Many existing FM-based approaches adopt generic architectures—such as convolutional neural networks

(CNNs) or transformers—that treat EEG or ECoG inputs as either flattened time-series or unordered feature vectors. This design overlooks the anatomical, spatial, and topological structure inherent in neural recordings, especially in sensor-based modalities like EEG, where the position and configuration of electrodes carry critical neurophysiological meaning (Craik et al. 2019).

To better accommodate the spatial structure of neural data, emerging models have begun to incorporate domain-specific architectural priors. Strategies include topographic positional encodings in transformers, atlas-based cortical projections, and graph representations using interchannel distances. However, a major unresolved challenge lies in the heterogeneity of recording setups. Electrode configurations vary widely across systems (e.g., 10–20 vs. high-density EEG), subjects, and clinical modalities like sEEG or ECoG—introducing structural variability that complicates spatial modeling and weakens cross-dataset generalization (Buriro et al. 2018; Chen, Chai, et al. 2024; Bahador et al. 2021).

To address this, spatial harmonization methods such as normalized scalp projections, Montreal Neurological Institute (MNI)-aligned cortical mappings, or 3D coordinate embeddings can provide a unified spatial basis. GNNs may benefit from adaptive edge construction mechanisms that learn connectivity directly from signal properties rather than relying on fixed physical layouts. Additionally, robust training strategies—such as masked channel modeling, spatial dropout, and domain-adaptive normalization—can improve model tolerance to inconsistent inputs and hardware-induced variability (Guo 2025; Cisetto et al. 2024; Wei et al. 2018).

Beyond spatial considerations, architectural designs should also reflect the multiscale nature of neural dynamics. The brain operates across a wide range of frequencies and temporal resolutions. Incorporating band-specific pathways, frequency-aware attention, wavelet-based encoders, or temporal dilations can enhance the model's ability to disentangle spectral features and capture both transient and sustained oscillatory components (Thangavel et al. 2021; Samavat et al. 2022).

Finally, structural adaptations should also account for intersubject variability. Techniques such as adaptive instance normalization, subject-specific modulation layers, or regularization based on subject identity can help preserve physiological individuality while learning robust shared representations. Taken together, these innovations point toward a new class of structure-aware neural architectures that balance anatomical grounding, cross-domain adaptability, and physiological relevance—advancing both the accuracy and translational potential of FM-based neural decoding (Saha and Baumert 2020; Pérez-Velasco et al. 2022; Fahimi et al. 2019).

To make the concept of spatial harmonization more explicit, recent approaches have explored several computational implementations. One effective method involves projecting electrode coordinates onto standardized template brains such as MNI space, enabling the alignment of electrode positions across subjects and hardware configurations. This harmonization facilitates cross-dataset training by providing a common spatial

reference that captures both anatomical and functional correspondence. Furthermore, GNNs operating on these normalized electrode graphs can explicitly model topological and functional connectivity patterns, allowing edges to represent either anatomical distances or dynamic coupling strength between sensors. Finally, embedding 3D electrode coordinates as positional encodings offers a lightweight yet powerful means of incorporating geometric priors directly into transformer-based attention layers, improving both spatial interpretability and intersubject generalization. Together, these techniques provide practical pathways for implementing structure-aware model architectures that respect the spatial organization of neural recordings while maintaining flexibility across heterogeneous datasets.

5.3 | Interpretability and Clinical Robustness

Interpretability is a prerequisite for deploying FM-based neural decoders in clinical and scientific domains. While attention mechanisms have become a common interpretability tool, their biological alignment remains limited, as attention weights often reflect computational saliency rather than known neural dynamics.

To address this limitation, more rigorous and neuroscientifically grounded interpretability methods should be incorporated throughout the model development pipeline. One such method is Layer-wise Relevance Propagation (LRP), which attributes a model's prediction back to its input by recursively propagating relevance scores through the network layers (Bach et al. 2015). In the context of neural signal decoding, LRP allows the decomposition of predictions into time, frequency, or spatial components—helping to identify when and where biologically meaningful signal features contribute to the decision. This is particularly useful in tasks such as seizure detection or ERP classification, where precise localization in time and space is clinically significant (Sturm et al. 2016; Borra and Magosso 2021).

Another powerful framework is Representational Similarity Analysis (RSA), which enables systematic comparison between internal model embeddings and known neural structures or cognitive states. By computing and comparing similarity matrices across neural activation patterns and model-derived representations, RSA evaluates whether the learned features align with canonical EEG rhythms, functional brain networks, or stimulus-response mappings—thereby providing a quantitative metric for biological plausibility (Kriegeskorte and Kievit 2013; Diedrichsen and Kriegeskorte 2017; Chen et al. 2013). Additionally, encoding–decoding alignment frameworks assess whether internal model representations can reconstruct meaningful behavioral, cognitive, or physiological variables. If latent features can accurately decode task conditions or re-encode known input stimuli, this supports their neuroscientific relevance (Caucheteux and King 2022; Gillioz 2024).

These methods are especially critical in high-stakes applications such as seizure prediction, mental state monitoring, or BCI control, where highlighting relevant frequency bands, cortical areas, or temporal epochs enhances diagnostic transparency and user trust. Beyond algorithmic techniques,

architectural strategies—such as hybrid models combining DL with symbolic reasoning or rule-based modules—may further improve interpretability by grounding representations in explicitly human-understandable logic (Craik et al. 2019; Samek et al. 2017).

Crucially, interpretability should not be treated as a post hoc visualization step. Instead, it must be integrated into the entire model development process—from the choice of training objectives and network architectures to evaluation protocols and visualization tools. Embedding interpretability at every stage is essential for fostering clinician trust, ensuring safety, and realizing the full translational potential of FM-based neural decoding in real-world scientific and clinical settings (Sadeghi et al. 2024).

6 | Enabling Conditions and System-Level Considerations for FM-Based Neural Decoding

The application of FMs to neural signal decoding holds significant potential, yet their transition from promising prototypes to reliable, real-world systems requires more than architectural innovation. Foundational challenges persist across multiple levels of the modeling pipeline—including data access and diversity, computational scalability, and the neurophysiological grounding of learned representations. This section outlines critical system-level considerations that must be addressed to enable the scalable, interpretable, and clinically viable deployment of FM-based neural decoders.

6.1 | Coordinated Data Integration and Privacy-Preserving Collaboration

A fundamental requirement for scalable FM-based neural decoding is access to large, diverse, and representative datasets. However, neural data are often fragmented across institutions due to privacy concerns, heterogeneous collection protocols, and logistical barriers. To address these challenges, collaborative data sharing and aggregation efforts are essential. Federated learning offers a promising privacy-preserving framework, enabling decentralized model training without raw data exchange. When combined with differential privacy or secure multiparty computation, such frameworks can facilitate ethically compliant collaboration while supporting population-level representation learning across clinical and research cohorts.

6.2 | Standardization and Harmonization of Heterogeneous

Another critical consideration is the harmonization of diverse data sources, particularly in terms of recording hardware (e.g., amplifier types, sampling rates, and channel configurations) and preprocessing pipelines (e.g., referencing and filtering). Without appropriate alignment, model performance may suffer due to domain shifts and inconsistent feature distributions. The adoption of standardized metadata schemas—such as the Brain Imaging Data Structure (BIDS) for EEG—as well as the

development of public benchmarking platforms with consistent evaluation protocols, are key to ensuring fair, reproducible, and generalizable model assessment.

6.3 | Biologically Valid Synthetic Data Augmentation

Simulation-based data augmentation provides a complementary path to increasing dataset diversity, especially for rare or underrepresented populations and conditions. Approaches such as data-driven signal synthesis, generative modeling, and neurophysiological simulation engines can expand the training corpus. However, to be effective, synthetic data must be carefully calibrated to preserve the statistical, spectral, and temporal characteristics of real neural recordings. Integration of such data should be accompanied by rigorous validation procedures to avoid introducing biologically implausible artifacts that could mislead model training.

6.4 | Real-Time Decoding and Continual Learning

For FM-based decoders to function in real-world BCI applications, they must move beyond offline inference and support real-time operation under nonstationary conditions. This necessitates both low-latency inference and continual adaptation. Lightweight transformer variants (e.g., Linformer, Performer, and MobileViT), in conjunction with model compression techniques such as pruning, quantization, and Low-Rank Adaptation (LoRA), offer a path toward efficient deployment on embedded or wearable devices. Meanwhile, continual learning strategies—including elastic weight consolidation, memory replay, and meta-learning—are critical for enabling adaptive decoding across sessions without catastrophic forgetting. While real-time adaptive FM decoding remains an open challenge, it is likely to be a central capability for future BCI systems operating in dynamic environments.

6.5 | Cross-Modality Challenges and Opportunities for Neural FMs

FMs for neural decoding have so far been developed predominantly in the context of EEG, owing to the broad availability of open-access datasets, standardized recording protocols, and relatively low barriers to large-scale data aggregation. However, extending FM paradigms across modalities such as ECoG and intracortical recordings presents both conceptual and practical challenges that are crucial for realizing the notion of truly unified neural representations.

6.5.1 | Signal Heterogeneity and Spatial Scale

EEG, ECoG, and intracortical recordings differ markedly in spatial resolution, signal amplitude, and sensitivity to noise. EEG provides whole-brain coverage but suffers from spatial blurring due to volume conduction, whereas ECoG captures mesoscopic cortical dynamics with improved spatial specificity. In contrast, intracortical microelectrode arrays yield high-resolution spiking activity but are limited to localized regions. These discrepancies

complicate direct cross-modal alignment of representations, as the same neural event may appear with vastly different temporal and spectral profiles across modalities. Effective cross-modal FMs must therefore learn scale-invariant features that generalize from macroscopic field potentials to fine-grained spiking dynamics.

6.5.2 | Data Imbalance and Standardization

Public EEG datasets number in the thousands of hours, while comparable large-scale ECoG or intracortical datasets remain scarce due to surgical and ethical constraints. The resulting imbalance poses difficulties for joint pretraining and may bias representations toward EEG-like characteristics. Harmonization initiatives that unify metadata, sampling schemes, and electrode localization standards (e.g., via MNI or CIFTI coordinate systems) are essential to enable consistent multimodal training.

6.5.3 | Architectural and Objective Adaptation

Architectures optimized for continuous EEG waveforms may be ill-suited for sparse, event-driven spiking data. To bridge this gap, future FMs could adopt hybrid architectures that integrate transformer-based sequence modeling with SNN modules, enabling the capture of both rate- and event-based dynamics within a shared framework. Similarly, self-supervised objectives may require modality-specific adjustments—such as masked voltage prediction for ECoG or spike-timing reconstruction for intracortical data—to ensure meaningful representation learning across signal types.

6.5.4 | Opportunities for Cross-Modal Transfer Learning

Despite these challenges, cross-modality pretraining offers promising opportunities. EEG-trained encoders may provide transferable initialization for ECoG models by leveraging shared oscillatory structure in cortical rhythms. Conversely, fine-tuned intracortical embeddings can inform population-level representations of microscale firing patterns, enhancing interpretability of macroscopic signals. Such transfer pathways could be realized through multiview contrastive learning or co-embedding frameworks that jointly align latent spaces across modalities.

6.5.5 | Toward Unified Neural FMs

Ultimately, achieving modality-agnostic representation learning will require the integration of heterogeneous neural datasets within a single pretraining framework. Combining EEG, ECoG, and intracortical signals under shared tokenization, normalization, and spatial alignment schemes could enable FMs that generalize across spatial scales—from population-level oscillations to neuronal spiking—thereby advancing both neuroscience research and clinical translation.

7 | Toward Clinical Translation

To bridge the gap between research-grade FMs and clinically validated diagnostic or therapeutic tools, it is crucial to establish a structured translational pathway that ensures reproducibility, reliability, and regulatory compliance. While recent advances have demonstrated the feasibility of large-scale neural FMs, their deployment in clinical practice remains constrained by the absence of standardized evaluation pipelines and insufficient integration with regulatory frameworks.

A clinically oriented translational roadmap may involve three progressive stages. First, multicenter and cross-institutional validation is essential to confirm generalizability across diverse patient populations, recording systems, and clinical conditions. Second, regulatory alignment must be pursued through established medical device pathways, such as the US Food and Drug Administration (FDA) guidelines for Software as a Medical Device (SaMD) and the Conformité Européenne (CE) marking processes under the European Union (EU) Medical Device Regulation (MDR). These processes require transparent performance reporting, documentation of model behavior, and consistent evaluation protocols. Third, post-deployment monitoring mechanisms should be implemented to detect data drift, hardware variability, or shifts in patient demographics that may affect model reliability over time. Such monitoring systems could include automated recalibration, clinical outcome auditing, and controlled continual learning strategies.

Achieving clinical-grade validation will require close interdisciplinary collaboration among clinicians, neuroscientists, engineers, and regulatory experts, alongside adherence to standardized guidelines for reproducibility and interpretability. This translational framework highlights that FMs for neural decoding should not only advance algorithmic performance but also align with the ethical, regulatory, and operational standards required for real-world deployment in neuroscience and medicine.

8 | Discussion

Neural signals provide a direct, high-resolution window into brain dynamics, making them a cornerstone of modern neuroscience and neurotechnology. However, their inherent complexity—marked by high dimensionality, strong nonstationarity, low SNRs, and substantial intersubject and intrasubject variability—continues to challenge traditional ML pipelines. Standard supervised approaches often struggle to generalize across recording setups or individuals and are further limited by their reliance on large quantities of labeled data and task-specific calibration.

FMs represent a promising shift in this landscape. By leveraging large-scale, heterogeneous, and often unlabeled neural datasets, FMs are capable of learning robust, transferable representations that generalize across individuals, sessions, and experimental paradigms. When guided by biologically aligned pretraining objectives, neurophysiologically informed architectural designs, and principled mechanisms for interpretability, FMs can address many of the structural limitations that have long constrained neural decoding.

Among their most transformative properties is the ability to support few- or zero-shot inference—enabling decoding in settings where labels are sparse or subject-specific adaptation is impractical. This has profound implications for real-time BCIs, passive neuro-monitoring, and population-level cognitive assessment.

Yet realizing this potential requires confronting several foundational challenges. Pretraining tasks must capture the non-linear, oscillatory, and spatially structured nature of brain activity. Architectures must accommodate variability in electrode configuration and incorporate domain-relevant priors across spatial and spectral dimensions. Interpretability must be treated as a first-class concern—not only for scientific transparency but also for clinical safety and trust. Moreover, these algorithmic advances must be underpinned by a robust data ecosystem: one that includes standardized formats, large-scale aggregation initiatives, simulation-based augmentation, and privacy-preserving collaborative frameworks such as federated learning.

While this review aimed to provide a broad perspective on FMs for neural signal decoding, it primarily focused on EEG-based approaches. This focus reflects the current landscape of neural data availability, as EEG remains the only modality with sufficiently large-scale, standardized, and publicly accessible datasets to enable FM-level pretraining and evaluation. Consequently, some of the proposed strategies—such as large-scale SSL or cross-subject generalization—have been discussed mainly within the EEG domain. Future studies should extend these frameworks to other modalities, including ECoG and intracortical recordings, as data resources, computational infrastructure, and multimodal harmonization initiatives continue to grow. Expanding such cross-modality efforts will be essential to achieving the overarching goal of unified neural representations.

Author Contributions

Youmin Shin and **Jii Kwon** jointly contributed to the conceptualization, methodology, formal analysis, investigation, and writing of the manuscript. **Youmin Shin** supervised the overall study, managed project administration, and provided resources. **Jii Kwon** primarily contributed to data curation and visualization. Both authors participated in the review and editing of the final manuscript and approved its submission.

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The authors declare no conflicts of interest.

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References

- Achiam, J., S. Adler, S. Agarwal, et al. 2023. “Gpt-4 Technical Report.” *arXiv Preprint arXiv*: 2303.08774.
- Agarwal, N., A. Mangal, S. Singiri, A. Chhapola, and S. Jain. 2022. “Self-Supervised Learning for EEG Artifact Detection.” Available at SSRN 4982677.
- Aksan, E., M. Kaufmann, P. Cao, and O. Hilliges. 2021. “A Spatio-Temporal Transformer for 3d Human Motion Prediction.” In *2021 International Conference on 3D Vision (3DV)*. IEEE.
- Aristimunya, B., D. Truong, P. Guetschel, et al. 2025. “EEG Foundation Challenge: From Cross-Task to Cross-Subject EEG Decoding.” *arXiv Preprint arXiv*: 2506.19141.
- Awais, M., M. Naseer, S. Khan, et al. 2025. “Foundation Models Defining a New Era in Vision: A Survey and Outlook.” *IEEE Transactions on Pattern Analysis and Machine Intelligence* 47: 2245–2264.
- Bach, S., A. Binder, G. Montavon, F. Klauschen, K. R. Müller, and W. Samek. 2015. “On Pixel-Wise Explanations for Non-Linear Classifier Decisions by Layer-Wise Relevance Propagation.” *PLoS ONE* 10, no. 7: e0130140.
- Bahador, N., J. Jokelainen, S. Mustola, and J. Kortelainen. 2021. “Reconstruction of Missing Channel in Electroencephalogram Using Spatiotemporal Correlation-Based Averaging.” *Journal of Neural Engineering* 18, no. 5: 056045.
- Banville, H., O. Chehab, A. Hyvärinen, D. A. Engemann, and A. Gramfort. 2021. “Uncovering the Structure of Clinical EEG Signals With Self-Supervised Learning.” *Journal of Neural Engineering* 18, no. 4: 046020.
- Barbano, C. A., M. Brunello, B. Dufumier, and M. Grangetto. 2024. “Anatomical Foundation Models for Brain MRIs.” *arXiv Preprint arXiv*: 2408.07079.
- Bethge, D. 2022. “Exploiting Multiple EEG Data Domains With Adversarial Learning.” In *2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*. IEEE.
- Bommasani, R. 2021. “On the Opportunities and Risks of Foundation Models.” *arXiv Preprint arXiv*: 2108.07258.
- Borra, D., and E. Magosso. 2021. “Deep Learning-Based EEG Analysis: Investigating P3 ERP Components.” *Journal of Integrative Neuroscience* 20, no. 4: 791–811.
- Brown, T., B. Mann, N. Ryder, et al. 2020. “Language Models Are Few-Shot Learners.” *Advances in Neural Information Processing Systems* 33: 1877–1901.
- Buriro, A. B., R. Shoorangiz, S. J. Weddell, and R. D. Jones. 2018. “Predicting Microsleep States Using EEG Inter-Channel Relationships.” *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 26, no. 12: 2260–2269.
- Buzsáki, G. 2004. “Large-Scale Recording of Neuronal Ensembles.” *Nature Neuroscience* 7, no. 5: 446–451.
- Buzsáki, G., C. A. Anastassiou, and C. Koch. 2012. “The Origin of Extracellular Fields and Currents—EEG, ECoG, LFP and Spikes.” *Nature Reviews Neuroscience* 13, no. 6: 407–420.
- Buzsáki, G., and A. Draguhn. 2004. “Neuronal Oscillations in Cortical Networks.” *Science* 304, no. 5679: 1926–1929.
- Caucheteux, C., and J.-R. King. 2022. “Brains and Algorithms Partially Converge in Natural Language Processing.” *Communications Biology* 5, no. 1: 134.

- Chefer, H., S. Gur, and L. Wolf. 2021. "Transformer Interpretability Beyond Attention Visualization." In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 782–791. IEEE.
- Chen, J., T. Feng, J. Shi, L. Liu, and H. Li. 2013. "Neural Representation of Decision Confidence." *Behavioural Brain Research* 245: 50–57.
- Chen, K., S. Chai, T. Xie, Q. Liu, and L. Ma. 2024. "EEG Spatial Inter-Channel Connectivity Analysis: A GCN-Based Dual Stream Approach to Distinguish Mental Fatigue Status." *Artificial Intelligence in Medicine* 157: 102996.
- Chen, R. J., T. Ding, M. Y. Lu, et al. 2024. "Towards a General-Purpose Foundation Model for Computational Pathology." *Nature Medicine* 30, no. 3: 850–862.
- Chen, X., C. Li, A. Liu, M. J. McKeown, R. Qian, and Z. J. Wang. 2022. "Toward Open-World Electroencephalogram Decoding via Deep Learning: A Comprehensive Survey." *IEEE Signal Processing Magazine* 39, no. 2: 117–134.
- Chen, Y., K. Ren, K. Song, et al. 2024. "EEGformer: Towards Transferable and Interpretable Large-Scale EEG Foundation Model." *arXiv Preprint arXiv*: 2401.10278.
- Choe, M., Y. Choi, J. Kwon, et al. 2025. "Increased Global and Regional Connectivity in Propofol-Induced Unconsciousness: Human Intracranial Electroencephalography Study." *Anesthesiology* 143, no. 1: 114–131.
- Cisotto, G., A. Zancanaro, I. F. Zoppis, and S. L. Manzoni. 2024. "hvEEGNet: A Novel Deep Learning Model for High-Fidelity EEG Reconstruction." *Frontiers in Neuroinformatics* 18: 1459970.
- Cox, J., P. Liu, S. E. Stolte, et al. 2024. "BrainSegFounder: Towards 3D Foundation Models for Neuroimage Segmentation." *Medical Image Analysis* 97: 103301.
- Craik, A., Y. He, and J. L. Contreras-Vidal. 2019. "Deep Learning for Electroencephalogram (EEG) Classification Tasks: A Review." *Journal of Neural Engineering* 16, no. 3: 031001.
- Cui, S., D. Lee, and D. Wen. 2024. "Toward Brain-Inspired Foundation Model for EEG Signal Processing: Our Opinion." *Frontiers in Neuroscience* 18: 1507654.
- Cui, W. 2024. "Neuro-GPT: Towards a Foundation Model for EEG." In *2024 IEEE International Symposium on Biomedical Imaging (ISBI)*. IEEE.
- Da Silva, C. P., S. Tedesco, and B. O'Flynn. 2024. "EEG Datasets for Healthcare: A Scoping Review." *IEEE Access* 12: 39186–39203.
- Dai, M., D. Zheng, R. Na, S. Wang, and S. Zhang. 2019. "EEG Classification of Motor Imagery Using a Novel Deep Learning Framework." *Sensors* 19, no. 3: 551.
- de Peralta, G., R. Menendez, Q. Noirhomme, et al. 2007. "Modern Electrophysiological Methods for Brain-Computer Interfaces." *Computational Intelligence and Neuroscience* 2007, no. 1: 056986.
- Défossez, A., L. Mazaré, M. Orsini, et al. 2024. "Moshi: A Speech-Text Foundation Model for Real-Time Dialogue." *arXiv Preprint arXiv*: 2410.00037.
- Del Pup, F., A. Zanola, L. F. Tshimanga, P. E. Mazzon, and M. Atzori. 2023. "SelfEEG: A Python Library for Self-Supervised Learning in Electroencephalography." *arXiv Preprint arXiv*: 2401.05405.
- Desai, S. A., T. Tcheng, and M. Morrell. 2019. "Transfer-Learning for Differentiating Epileptic Patients Who Respond to Treatment Based on Chronic Ambulatory ECoG Data." In *2019 9th International IEEE/EMBS Conference on Neural Engineering (NER)*. IEEE.
- Diedrichsen, J., and N. Kriegeskorte. 2017. "Representational Models: A Common Framework for Understanding Encoding, Pattern-Component, and Representational-Similarity Analysis." *PLoS Computational Biology* 13, no. 4: e1005508.
- Doshi-Velez, F., and B. Kim. 2017. "Towards a Rigorous Science of Interpretable Machine Learning." *arXiv Preprint arXiv*: 1702.08608.
- Dwivedi, V. P., V. Schlegel, A. T. Liu, et al. 2024. "Representation Learning of Structured Data for Medical Foundation Models." *arXiv Preprint arXiv*: 2410.13351.
- Eldele, E., Z. Chen, C. Liu, et al. 2021. "An Attention-Based Deep Learning Approach for Sleep Stage Classification With Single-Channel EEG." *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 29: 809–818.
- Fahimi, F., Z. Zhang, W. B. Goh, T. S. Lee, K. K. Ang, and C. Guan. 2019. "Inter-Subject Transfer Learning With an End-to-End Deep Convolutional Neural Network for EEG-Based BCI." *Journal of Neural Engineering* 16, no. 2: 026007.
- Farahat, A., C. Reichert, C. M. Sweeney-Reed, and H. Hinrichs. 2019. "Convolutional Neural Networks for Decoding of Covert Attention Focus and Saliency Maps for EEG Feature Visualization." *Journal of Neural Engineering* 16, no. 6: 066010.
- Gillioz, V. 2024. "Alignment of Large Language Models and Brain Activity: Exploring Language Processing Through sEEG in a Multimodal Syntactic Task."
- Goldberger, A. L., L. A. N. Amaral, L. Glass, et al. 2000. "PhysioBank, PhysioToolkit, and PhysioNet: Components of a New Research Resource for Complex Physiologic Signals." *Circulation* 101, no. 23: e215–e220.
- Graimann, B. 2005. "A Comparison Between Using ECoG and EEG for Direct Brain Communication. Proceedings of the EMBEC05."
- Gramfort, A. 2021. "Learning With Self-Supervision on EEG Data." In *2021 9th International Winter Conference on Brain-Computer Interface (BCI)*. IEEE.
- Grigorovsky, V., D. Jacobs, V. L. Breton, et al. 2020. "Delta-Gamma Phase-Amplitude Coupling as a Biomarker of Postictal Generalized EEG Suppression." *Brain Communications* 2, no. 2: fcaa182.
- Gundu, S. R., and C. A. Panem. 2025. "A Review on Contemporary Brain-Computer Interface Researches and Limitations." *Brain-Computer Interfaces, Advances in Neural Engineering*: 287–295 Elsevier.
- Guo, H.-H., Y. Hu, K. Liu, et al. 2024. "Firededts: A Foundation Text-to-Speech Framework for Industry-Level Generative Speech Applications." *arXiv Preprint arXiv*: 2409.03283.
- Guo, Z. 2025. "Neuro-3D: Towards 3D Visual Decoding From EEG Signals. In Proceedings of the Computer Vision and Pattern Recognition Conference."
- Gurmessa, D. K., and W. Jimma. 2025. "A Comprehensive Evaluation of Interpretable Artificial Intelligence for Epileptic Seizure Diagnosis Using an Electroencephalogram: A Systematic Review." *DIGITAL HEALTH* 11: 20552076251325411.
- Han, D.-K., and J.-H. Jeong. 2021. "Domain Generalization for Session-Independent Brain-Computer Interface." In *2021 9th International Winter Conference on Brain-Computer Interface (BCI)*. IEEE.
- Han, J., X. Wei, and A. A. Faisal. 2023. "EEG Decoding for Datasets With Heterogenous Electrode Configurations Using Transfer Learning Graph Neural Networks." *Journal of Neural Engineering* 20, no. 6: 66027.
- Hayati, H. 2025. "Wireless Data and Power Transfer Solutions for High-Density Neural Electrophysiology in Live Animal Models."
- Hutson, M. 2018. "Artificial Intelligence Faces Reproducibility Crisis." *Science* 359: 725–726.
- Jayaram, V., and A. Barachant. 2018. "MOABB: Trustworthy Algorithm Benchmarking for BCIs." *Journal of Neural Engineering* 15, no. 6: 066011.

- Jiang, W.-B., Y. Wang, B. L. Lu, and D. Li. 2024. "NeuroLM: A Universal Multi-Task Foundation Model for Bridging the Gap Between Language and EEG Signals." *arXiv Preprint arXiv*: 2409.00101.
- Jiang, W.-B., L.-M. Zhao, and B.-L. Lu. 2024. "Large Brain Model for Learning Generic Representations With Tremendous EEG Data in BCI." *arXiv Preprint arXiv*: 2405.18765.
- Joshi, A., P. S. Matharu, L. Malviya, M. Kumar, and A. Jadhav. 2025. "Advancing EEG Based Stress Detection Using Spiking Neural Networks and Convolutional Spiking Neural Networks." *Scientific Reports* 15, no. 1: 26267.
- Keene, D. L., S. Whiting, and E. C. Ventureyra. 2000. "Electrocorticography." *Epileptic Disorders* 2, no. 1: 57–63.
- Klein, T., P. Minakowski, and S. Sager. 2025. "Flexible Patched Brain Transformer Model for EEG Decoding." *Scientific Reports* 15, no. 1: 1–12.
- Kriegeskorte, N., and R. A. Kievit. 2013. "Representational Geometry: Integrating Cognition, Computation, and the Brain." *Trends in Cognitive Sciences* 17, no. 8: 401–412.
- Ku, P.-J., A. H. Liu, R. Korostik, S. F. Huang, S. W. Fu, and A. Jukić. 2025. "Generative Speech Foundation Model Pretraining for High-Quality Speech Extraction and Restoration." In *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE.
- Lehembre, R., O. Gosseries, Z. Lugo, et al. 2012. "Electrophysiological Investigations of Brain Function in Coma, Vegetative and Minimally Conscious Patients." *Archives Italiennes de Biologie* 150, no. 2-3: 122–139.
- Lewis, C. M., A. Hoffmann, and F. Helmchen. 2024. "Linking Brain Activity Across Scales With Simultaneous Opto- and Electrophysiology." *Neurophotonics* 11, no. 3: 033403.
- Li, C., Y. Qi, X. Ding, J. Zhao, T. Sang, and M. Lee. 2022. "A Deep Learning Method Approach for Sleep Stage Classification With EEG Spectrogram." *International Journal of Environmental Research and Public Health* 19, no. 10: 6322.
- Liu, F., D. Chen, Z. Guan, et al. 2024. "Remoteclick: A Vision Language Foundation Model for Remote Sensing." *IEEE Transactions on Geoscience and Remote Sensing* 62: 1–16.
- Lopez-Bernal, D., D. Balderas, P. Ponce, and A. Molina. 2024. "Exploring Inter-Trial Coherence for Inner Speech Classification in EEG-Based Brain–Computer Interface." *Journal of Neural Engineering* 21, no. 2: 026048.
- Lotte, F., L. Bougrain, A. Cichocki, et al. 2018. "A Review of Classification Algorithms for EEG-Based Brain–Computer Interfaces: A 10 Year Update." *Journal of Neural Engineering* 15, no. 3: 031005.
- Mak, J. N., and J. R. Wolpaw. 2009. "Clinical Applications of Brain-Computer Interfaces: Current State and Future Prospects." *IEEE Reviews in Biomedical Engineering* 2: 187–199.
- Moon, H., J. Kwon, J. Eun, et al. 2024. "Electrocorticogram (ECoG): Engineering Approaches and Clinical Challenges for Translational Medicine." *Advanced Materials Technologies* 9, no. 12: 2301692.
- Morlet, D., J. Mattout, C. Fischer, et al. 2023. "Intraclinical Detection of Voluntary Attention in Coma and Post-Coma Patients Using Electrophysiology." *Clinical Neurophysiology* 145: 151–161.
- Mumtaz, W., S. Rasheed, and A. Irfan. 2021. "Review of Challenges Associated With the EEG Artifact Removal Methods." *Biomedical Signal Processing and Control* 68: 102741.
- Myers, D., R. Mohawesh, V. I. Chellaboina, et al. 2024. "Foundation and Large Language Models: Fundamentals, Challenges, Opportunities, and Social Impacts." *Cluster Computing* 27, no. 1: 1–26.
- Narayan, A., I. Chami, L. Orr, S. Arora, and C. Ré. 2022. "Can Foundation Models Wrangle Your Data?" *arXiv Preprint arXiv*: 2205.09911.
- Obeid, I., and J. Picone. 2016. "The Temple University Hospital EEG Data Corpus." *Frontiers in Neuroscience* 10: 196.
- Onslow, A. C., R. Bogacz, and M. W. Jones. 2011. "Quantifying Phase-Amplitude Coupling in Neuronal Network Oscillations." *Progress in Biophysics and Molecular Biology* 105, no. 1–2: 49–57.
- Paaß, G., and S. Giesselbach. 2023. *Foundation Models for Natural Language Processing: Pre-Trained Language Models Integrating Media*. Springer Nature.
- Pérez-Velasco, S., E. Santamaria-Vazquez, V. Martinez-Cagigal, D. Marcos-Martinez, and R. Hornero. 2022. "EEGSym: Overcoming Inter-Subject Variability in Motor Imagery Based BCIs With Deep Learning." *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 30: 1766–1775.
- Peterson, S. M., Z. Steine-Hanson, N. Davis, R. P. N. Rao, and B. W. Brunton. 2021. "Generalized Neural Decoders for Transfer Learning Across Participants and Recording Modalities." *Journal of Neural Engineering* 18, no. 2: 026014.
- Priyasad, D., T. Fernando, S. Denman, S. Sridharan, and C. Fookes. 2022. "Affect Recognition From Scalp-EEG Using Channel-Wise Encoder Networks Coupled With Geometric Deep Learning and Multi-Channel Feature Fusion." *Knowledge-Based Systems* 250: 109038.
- Roy, Y., H. Banville, I. Albuquerque, A. Gramfort, T. H. Falk, and J. Faubert. 2019. "Deep Learning-Based Electroencephalography Analysis: A Systematic Review." *Journal of Neural Engineering* 16, no. 5: 051001.
- Rudin, C. 2019. "Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead." *Nature Machine Intelligence* 1, no. 5: 206–215.
- Sadeghi, Z., R. Alizadehsani, M. A. Cifci, et al. 2024. "A Review of Explainable Artificial Intelligence in Healthcare." *Computers and Electrical Engineering* 118: 109370.
- Saha, S., and M. Baumert. 2020. "Intra-and Inter-Subject Variability in EEG-Based Sensorimotor Brain–Computer Interface: A Review." *Frontiers in Computational Neuroscience* 13: 87.
- Samavat, A., E. Khalili, B. Ayati, and M. Ayati. 2022. "Deep Learning Model With Adaptive Regularization for EEG-Based Emotion Recognition Using Temporal and Frequency Features." *IEEE Access* 10: 24520–24527.
- Samek, W., T. Wiegand, and K.-R. Müller. 2017. "Explainable Artificial Intelligence: Understanding, Visualizing and Interpreting Deep Learning Models." *arXiv Preprint arXiv*: 1708.08296.
- Samiee, S., and S. Baillet. 2017. "Time-Resolved Phase-Amplitude Coupling in Neural Oscillations." *NeuroImage* 159: 270–279.
- Schirrmeister, R. T., J. T. Springenberg, L. D. J. Fiederer, et al. 2017. "Deep Learning With Convolutional Neural Networks for EEG Decoding and Visualization." *Human Brain Mapping* 38, no. 11: 5391–5420.
- Shende, P. M., and V. S. Jabade. 2015. "Literature Review of Brain Computer Interface (BCI) Using Electroencephalogram Signal." In *2015 International Conference on Pervasive Computing (ICPC)*. IEEE.
- Shi, E., K. Zhao, Q. Yuan, et al. 2024. "Fome: A Foundation Model for EEG Using Adaptive Temporal-Lateral Attention Scaling." *arXiv Preprint arXiv*: 2409.12454.
- Shimizu, D., K. Watanabe, and A. Dengel. 2024. "Feature Estimation of Global Language Processing in EEG Using Attention Maps." In *Proceedings of the Asian Conference on Computer Vision (ACCV)*, 1706–1722. Springer.

- Shin, Y., J. S. Kim, and C. K. Chung. 2021. "Proof of Beat Entrainment and Its Characteristics." In *2021 9th International Winter Conference on Brain-Computer Interface (BCI)*. IEEE.
- Shin, Y., J. Kwon, J. S. Kim, and C. K. Chung. 2022. "Introduction of Beat Oscillation to Improve the Performance of Music BCI Decoder." In *2022 10th International Winter Conference on Brain-Computer Interface (BCI)*. IEEE.
- Simeral, J. D., T. Hosman, J. Saab, et al. 2021. "Home Use of a Percutaneous Wireless Intracortical Brain-Computer Interface by Individuals With Tetraplegia." *IEEE Transactions on Biomedical Engineering* 68, no. 7: 2313–2325.
- Staba, R. J., M. Stead, and G. A. Worrell. 2014. "Electrophysiological Biomarkers of Epilepsy." *Neurotherapeutics* 11, no. 2: 334–346.
- Stevens, S. 2024. "Bioclip: A Vision Foundation Model for the Tree of Life." In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE.
- Sturm, I., S. Lapuschkin, W. Samek, and K. R. Müller. 2016. "Interpretable Deep Neural Networks for Single-Trial EEG Classification." *Journal of Neuroscience Methods* 274: 141–145.
- Sujatha Ravindran, A., and J. Contreras-Vidal. 2023. "An Empirical Comparison of Deep Learning Explainability Approaches for EEG Using Simulated Ground Truth." *Scientific Reports* 13, no. 1: 17709.
- Sun, Y., L. Wang, G. Li, W. Lin, and L. Wang. 2025. "A Foundation Model for Enhancing Magnetic Resonance Images and Downstream Segmentation, Registration and Diagnostic Tasks." *Nature Biomedical Engineering* 9, no. 4: 521–538.
- Suzumura, T., H. Kanezashi, and S. Akahori. 2024. "Graph Adapter of EEG Foundation Models for Parameter Efficient Fine Tuning." *arXiv Preprint arXiv*: 2411.16155.
- Tabar, Y. R., and U. Halici. 2016. "A Novel Deep Learning Approach for Classification of EEG Motor Imagery Signals." *Journal of Neural Engineering* 14, no. 1: 016003.
- Tak, D., B. A. Garomsa, T. L. Chaunzwa, et al. 2024. "A Foundation Model for Generalized Brain MRI Analysis." *Medrxiv*.
- Thangavel, P., J. Thomas, W. Y. Peh, et al. 2021. "Time–Frequency Decomposition of Scalp Electroencephalograms Improves Deep Learning-Based Epilepsy Diagnosis." *International Journal of Neural Systems* 31, no. 8: 2150032.
- Thomas, A. W., H. R. Heekeren, K. R. Müller, and W. Samek. 2019. "Analyzing Neuroimaging Data Through Recurrent Deep Learning Models." *Frontiers in Neuroscience* 13: 1321.
- Torres, J. M. M., S. Medina-DeVilliers, T. Clarkson, M. D. Lerner, and G. Riccardi. 2023. "Evaluation of Interpretability for Deep Learning Algorithms in EEG Emotion Recognition: A Case Study in Autism." *Artificial Intelligence in Medicine* 143: 102545.
- Tortora, S., S. Ghidoni, C. Chisari, S. Micera, and F. Artoni. 2020. "Deep Learning-Based BCI for Gait Decoding From EEG With LSTM Recurrent Neural Network." *Journal of Neural Engineering* 17, no. 4: 046011.
- Tripathi, M., P. K. Singh, D. Vibha, et al. 2010. "Electrophysiological Characteristics of Seizure Clusters." *Clinical EEG and Neuroscience* 41, no. 3: 143–146.
- Truong, D., M. A. Khalid, and A. Delorme. 2024. "EEG-SSL: A Framework for Self-Supervised Learning on EEG." In *2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*. IEEE.
- Turgut, Ö., F. S. Bott, M. Ploner, and D. Rueckert. 2025. "Are Foundation Models Useful Feature Extractors for Electroencephalography Analysis?" *arXiv Preprint arXiv*: 2502.21086.
- Ullah, I., M. Hussain, and H. Aboalsamh. 2018. "An Automated System for Epilepsy Detection Using EEG Brain Signals Based on Deep Learning Approach." *Expert Systems With Applications* 107: 61–71.
- Vallabhaneni, R. B., P. Sharma, V. Kumar, et al. 2021. "Deep Learning Algorithms in EEG Signal Decoding Application: A Review." *IEEE Access* 9: 125778–125786.
- Waldert, S. 2016. "Invasive vs. Non-Invasive Neuronal Signals for Brain-Machine Interfaces: Will One Prevail?" *Frontiers in Neuroscience* 10: 295.
- Wang, E. Y., P. G. Fahey, Z. Ding, et al. 2025. "Foundation Model of Neural Activity Predicts Response to New Stimulus Types." *Nature* 640, no. 8058: 470–477.
- Wang, J., S. Zhao, Z. Luo, et al. 2024. "Cbramod: A Criss-Cross Brain Foundation Model for EEG Decoding." *arXiv Preprint arXiv*: 2412.07236.
- Wang, L., T. Suzumura, and H. Kanezashi. 2024. "Graph-Enhanced EEG Foundation Model." *arXiv Preprint arXiv*: 2411.19507.
- Wang, M., W. Han, I. Shafran, et al. 2023. "Slm: Bridge the Thin Gap Between Speech and Text Foundation Models." In *2023 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*. IEEE.
- Wang, X.-J. 2010. "Neurophysiological and Computational Principles of Cortical Rhythms in Cognition." *Physiological Reviews* 90, no. 3: 1195–1268.
- Wang, X.-W., D. Nie, and B.-L. Lu. 2014. "Emotional State Classification From EEG Data Using Machine Learning Approach." *Neurocomputing* 129: 94–106.
- Wang, Y., W. Fan, S. Wang, and Y. Ma. 2025. "Towards Graph Foundation Models: A Transferability Perspective." *arXiv Preprint arXiv*: 2503.09363.
- Wang, Y., N. Huang, N. Mammone, M. Cecchi, and X. Zhang. 2025. "LEAD: Large Foundation Model for EEG-Based Alzheimer's Disease Detection." *arXiv Preprint arXiv*: 2502.01678.
- Wei, X., L. Zhou, Z. Chen, L. Zhang, and Y. Zhou. 2018. "Automatic Seizure Detection Using Three-Dimensional CNN Based on Multi-Channel EEG." *BMC Medical Informatics and Decision Making* 18, no. Suppl 5: 111.
- Weng, W., Y. Gu, S. Guo, et al. 2024. "Self-Supervised Learning for Electroencephalogram: A Systematic Survey." *ACM Computing Surveys* 57, no. 12: 1–38. ACM.
- Wu, D., Y. Xu, and B.-L. Lu. 2020. "Transfer Learning for EEG-Based Brain–Computer Interfaces: A Review of Progress Made Since 2016." *IEEE Transactions on Cognitive and Developmental Systems* 14, no. 1: 4–19.
- Wu, J. 2024. "General Object Foundation Model for Images and Videos at Scale." In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 3783–3795. IEEE.
- Xu, H., Q. Ye, M. Yan, et al. 2023. "Mplug-2: A Modularized Multi-Modal Foundation Model Across Text, Image and Video." In *International Conference on Machine Learning*. PMLR.
- Xu, L., M. Xu, Y. Ke, X. An, S. Liu, and D. Ming. 2020. "Cross-Dataset Variability Problem in EEG Decoding With Deep Learning." *Frontiers in Human Neuroscience* 14: 103.
- Yang, J., Z. Ma, J. Wang, and Y. Fu. 2020. "A Novel Deep Learning Scheme for Motor Imagery EEG Decoding Based on Spatial Representation Fusion." *IEEE Access* 8: 202100–202110.
- Yang, Z., X. Xu, J. Zhang, G. Wang, M. K. Kalra, and P. Yan. 2025. "Chest X-Ray Foundation Model With Global and Local Representations Integration." *arXiv Preprint arXiv*: 2502.05142.
- Yuan, L., D. Chen, Y. L. Chen, et al. 2021. "Florence: A New Foundation Model for Computer Vision." *arXiv Preprint arXiv*: 2111.11432.

- Yuan, Y. 2023. "On the Power of Foundation Models." In *International Conference on Machine Learning*. PMLR.
- Yuan, Y., G. Xun, K. Jia, and A. Zhang. 2018. "A Multi-View Deep Learning Framework for EEG Seizure Detection." *IEEE Journal of Biomedical and Health Informatics* 23, no. 1: 83–94.
- Yue, T., S. Xue, X. Gao, et al. 2024. "BrainGPT: Unleashing the Potential of EEG Generalist Foundation Model by Autoregressive Pre-Training." *arXiv Preprint arXiv*: 2410.19779.
- Zeng, H., C. Yang, G. Dai, F. Qin, J. Zhang, and W. Kong. 2018. "EEG Classification of Driver Mental States by Deep Learning." *Cognitive Neurodynamics* 12, no. 6: 597–606.
- Zhang, X., N. Ou, B. Doga Basaran, et al. 2025. "A Foundation Model for Lesion Segmentation on Brain MRI With Mixture of Modality Experts." *IEEE Transactions on Medical Imaging* 44: 2594–2604.
- Zhang, X., Z. Zhao, T. Tsiligkaridis, and M. Zitnik. 2022. "Self-Supervised Contrastive Pre-Training for Time Series via Time-Frequency Consistency." *Advances in Neural Information Processing Systems* 35: 3988–4003.
- Zhang, Z., J. M. Fort, and L. Giménez Mateu. 2024. "Mini Review: Challenges in EEG Emotion Recognition." *Frontiers in Psychology* 14: 1289816.
- Zhou, X., C. Liu, Z. Chen, et al. 2025. "Brain Foundation Models: A Survey on Advancements in Neural Signal Processing and Brain Discovery." *arXiv Preprint arXiv*: 2503.00580.